Linear Algebra II (2026 Fall)

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Chapter 6. Inner Product Spaces

- Section 6.2 : The Gram–Schmidt orthogonalization process and orthogonal complements

Section 6.2. The Gram-Schmidt orthogonalization process and orthogonal complements

Recall : Let V be a vector space over a field F.

(1) An inner product on V is a function

$$\langle \cdot , \cdot \rangle : V \times V \to F$$

which satisfies the following conditions:

(a) $\langle x+z, y\rangle = \langle x, y\rangle + \langle z, y\rangle$.
(b) $\langle cx, y\rangle = c\langle x, y\rangle$.
(c) $\overline{\langle x, y\rangle} = \langle y, x\rangle$, *where the bar denotes complex conjugation.*
(d) $\langle x, x\rangle > 0$ *if* $x \neq 0$.

(2) Let $S = \{v_1, v_2, \ldots\}$ be a subset of V. We say that S is orthonormal if

$\langle v_i, v_j \rangle = \delta_{ij}$ for all i, j, where δ_{ij} denote the Kronecker delta.

(3) For $\mathbb{R}^n$ and $\mathbb{C}^n$, the standard ordered bases are orthonormal.

Definition : Let V be an inner product space.

An ordered basis for V is called an orthonormal basis if it is orthonormal.

Example 2. : The following set is an orthonormal basis for $\mathbb{R}^2$:

$$\left\{\left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right), \left(\frac{2}{\sqrt{5}}, \frac{-1}{\sqrt{5}}\right)\right\}$$

Remark : Let V be an inner product space, and let $\beta = \{v_1, v_2, \ldots, v_n\}$ be a basis for V.

(1) For $x = a_1 v_1 + \cdots + a_n v_n, y = b_1 v_1 + \cdots + b_n v_n \in V$, we have

$$\langle x, y \rangle = \sum_{i,j} a_i \overline{b_j} \langle v_i, v_j \rangle.$$

In particular, $\langle x, y \rangle$ is determined by

$$[x]_\beta = (a_1, \ldots, a_n), \ [y]_\beta = (b_1, \ldots, b_n) \text{ and } \{ \langle v_i, v_j \rangle \mid 1 \le i, j \le n \}.$$

(2) Suppose that β is an orthonormal basis for V. Then $\langle v_i, v_j \rangle = \delta_{ij}$ and hence

$$\langle x, y \rangle = \sum_{i=1}^{n} a_i \overline{b_i} \ .$$

The next theorem and its corollaries will illustrate

the importance of orthonormal sets and orthonormal bases.

Theorem 6.3. : Let V be an inner product space, and

let $S = \{v_1, v_2, \ldots, v_k\}$ be an orthogonal subset of V consisting of nonzero vectors.

If $y \in \text{span}(S)$, then

$$y = \sum_{i=1}^{k} \frac{\langle y, v_i \rangle}{\|v_i\|^2} v_i.$$

Proof : Write $y = a_1 v_1 + a_2 v_2 + \cdots + a_k v_k$.

Then

$$\langle y, v_j \rangle = \sum_{i=1}^{k} a_i \langle v_i, v_j \rangle = a_j \langle v_j, v_j \rangle = a_j \|v_j\|^2.$$

Thus it holds that $a_j = \dfrac{\langle y, v_j \rangle}{\|v_j\|^2}$ for all $1 \le j \le k$. ■

Corollary 1. : Let V be an inner product space, and

let $S = \{v_1, v_2, \ldots, v_k\}$ be an orthonormal subset of V.

If $y \in \text{span}(S)$, then

$$y = \sum_{i=1}^{k} \langle y, v_i \rangle v_i.$$

Corollary 2. : Let V be an inner product space, and

let $S = \{v_1, v_2, \ldots, v_k\}$ be an orthogonal subset of V consisting of nonzero vectors.

Then S is linearly independent.

Proof : Suppose that $a_1 v_1 + \cdots + a_k v_k = O_V$. Then

$$\langle a_1 v_1 + \cdots + a_k v_k, v_i \rangle = \langle O_V, v_i \rangle = 0 \text{ for all } 1 \le i \le k,$$

and hence $a_i \|v_i\|^2 = 0$.

This implies that $a_i = 0$ for all $1 \le i \le k$ since S consists of nonzero vectors.

Example 3. : Recall the following example in the previous section.

Example 8. : The subset $S = \{u=(1,1,0), v=(1,-1,1), w=(-1,1,2)\}$ of F^3 is orthogonal.

Also $\|u\| = \sqrt{2}$, $\|v\| = \sqrt{3}$ and $\|w\| = \sqrt{6}$, and hence S is not orthonormal.

By normalizing process, we obtain the following orthonormal set:

$$\bar{S} = \left\{\bar{u} = \frac{1}{\sqrt{2}}(1,1,0), \bar{v} = \frac{1}{\sqrt{3}}(1,-1,1), \bar{w} = \frac{1}{\sqrt{6}}(-1,1,2)\right\}$$

Thus $\bar{S}$ is linearly independent by Corollary 2.

Since $\mathbb{R}^3$ is of dimension 3, it follows that $\bar{S}$ is an orthonormal basis for $\mathbb{R}^3$.

Now, we can easily express $x = (2,1,3)$ by the vectors in $\bar{S}$.

Indeed, $\langle x, \bar{u}\rangle = \frac{3}{\sqrt{2}}$, $\langle x, \bar{v}\rangle = \frac{4}{\sqrt{3}}$ and $\langle x, \bar{w}\rangle = \frac{5}{\sqrt{6}}$.

Thus we have

$$x = \frac{3}{2}(1,1,0) + \frac{4}{3}(1,-1,1) + \frac{5}{6}(-1,1,2).$$

◎ The Gram-Schmidt process

From now on, we study the problem how to find an orthonormal basis

when an ordered basis for a finite dimensional inner product space is given.

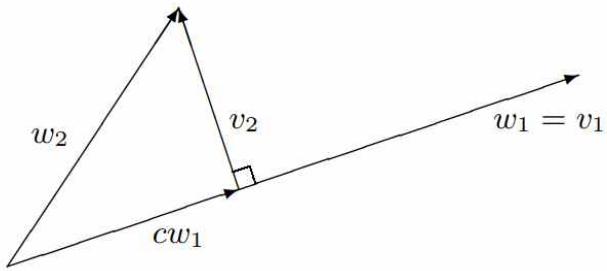
Example : Let $\{w_1, w_2\}$ be a basis for $\mathbb{R}^2$.

We will find a scalar $c \in \mathbb{R}$ such that w_1 and $w_2 - cw_1$ is orthogonal.

Indeed, the desired condition $\langle w_1, w_2 - cw_1 \rangle = \langle w_1, w_2 \rangle - c\|w_1\|^2 = 0$ is satisfied if

$$c = \frac{\langle w_1, w_2 \rangle}{\|w_1\|^2}.$$

Letting $v_1 = w_1$ and $v_2 = w_2 - \dfrac{\langle w_1, w_2 \rangle}{\|w_1\|^2} w_1$, it follows that $\{v_1, v_2\}$ is orthogonal.



Finally, by the normalizing process, we get the following orthonormal basis for $\mathbb{R}^2$:

$$\left\{ \frac{1}{\|v_1\|} v_1, \frac{1}{\|v_2\|} v_2 \right\}$$

Theorem 6.4. : Let V be an inner product space, and let $S = \{w_1, w_2, \ldots, w_n\}$ be a linearly independent subset of V. Define $S' = \{v_1, v_2, \ldots, v_n\}$, where

$$v_1 = w_1$$

and

$$v_k = w_k - \sum_{j=1}^{k-1} \frac{\langle w_k, v_j \rangle}{\|v_j\|^2} v_j \text{ for } 2 \le k \le n.$$

Then S' is an orthogonal set of nonzero vectors such that $\text{span}(S) = \text{span}(S')$.

Proof : We use induction on $n \ge 2$.

If $n = 2$ and so $S = \{w_1, w_2\}$, then

$S' = \left\{ v_1, v_2 = w_2 - \frac{\langle w_1, w_2 \rangle}{\|w_1\|^2} w_1 \right\}$ is an orthogonal set of nonzero vectors and $\text{span}(S) = \text{span}(S')$.

Suppose that $n \ge 3$, and consider $S_0 = \{w_1, w_2, \ldots, w_{n-1}\}$ and $S_0' = \{v_1, v_2, \ldots, v_{n-1}\}$.

By induction hypothesis, S_0' is an orthogonal set of nonzero vectors and $\text{span}(S_0) = \text{span}(S_0')$.

Also, $v_n = w_n - \sum_{j=1}^{n-1} \frac{\langle w_n, v_j \rangle}{\|v_j\|^2} v_j$.

Proof (Continued) :

If $v_n = O_V$, then $w_n = \sum_{j=1}^{n-1} \frac{\langle w_n, v_j \rangle}{\|v_j\|^2} v_j \in \text{span}(S_0') = \text{span}(S_0)$.

This is a contradiction since $S = \{w_1, w_2, \ldots, w_n\}$ is linearly independent. Thus $v_n \neq O_V$.

Next, consider $\langle v_i, v_j \rangle$ for $1 \leq i < j \leq n$.

If $j \leq n-1$, then $\langle v_i, v_j \rangle = 0$ since S_0' is orthogonal.

Also, if $j = n$ then we have

$$\langle v_i, v_n \rangle = \left\langle v_i, w_n - \sum_{j=1}^{n-1} \frac{\langle w_n, v_j \rangle}{\|v_j\|^2} v_j \right\rangle = \langle v_i, w_n \rangle - \frac{\overline{\langle w_n, v_j \rangle}}{\|v_j\|^2} \langle v_i, v_i \rangle$$

$$= \langle v_i, w_n \rangle - \overline{\langle w_n, v_j \rangle} = 0.$$

This completes the proof that S' is an orthogonal set of nonzero vectors.

Finally, we have

$$\text{span}(S) = \text{span}(S_0 \cup \{w_n\}) = \text{span}(S_0' \cup \{w_n\}) = \text{span}(S_0' \cup \{v_n\}) = \text{span}(S'). \quad \blacksquare$$

Remark : The construction of $S' = \{v_1, v_2, \ldots, v_n\}$ by the use of Theorem 6.4 is called the Gram-Schmidt process.

Example 4. : Let $S := \{w_1 = (1,0,1,0), w_2 = (1,1,1,1), w_3 = (0,1,2,1)\} \subset \mathbb{R}^4$.

(1) S is linearly independent.

(2) (the Gram-Schmidt process)

Step 1. : $v_1 = w_1 = (1,0,1,0)$

Step 2. : $v_2 = w_2 - \frac{\langle w_2, v_1 \rangle}{\|v_1\|^2} v_1 = (1,1,1,1) - \frac{2}{2}(1,0,1,0) = (0,1,0,1)$

Step 3. : $v_3 = w_3 - \frac{\langle w_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle w_3, v_2 \rangle}{\|v_2\|^2} v_2 = (0,1,2,1) - \frac{2}{2}(1,0,1,0) - \frac{2}{2}(0,1,0,1) = (-1,0,1,0)$

Thus $\{v_1, v_2, v_3\}$ is an orthogonal basis for $\mathrm{span}(S)$.

(3) (the normalization process)

$$u_1 = \frac{1}{\|v_1\|} v_1 = \frac{1}{\sqrt{2}}(1,0,1,0), \quad u_2 = \frac{1}{\|v_2\|} v_2 = \frac{1}{\sqrt{2}}(0,1,0,1), \quad u_3 = \frac{1}{\| \, \| v_3 \| \, \|} v_3 = \frac{1}{\sqrt{2}}(-1,0,1,0)$$

Thus $\{u_1, u_2, u_3\}$ is an orthonormal basis for $\mathrm{span}(S)$.

Example 5. : Let $V = P(\mathbb{R})$ with the inner product

$$\langle f(x), g(x) \rangle = \int_{-1}^{1} f(t)g(t)dt.$$

(1) $P_2(\mathbb{R})$ is a subspace of V and $\beta = \{1, x, x^2\}$ is the standard ordered basis for $P_2(\mathbb{R})$.

(2) (the Gram-Schmidt process)

Step 1. : $v_1 = 1$

Step 2. : $v_2 = x - \frac{\langle x,1 \rangle}{\|1\|^2} 1 = x$

Step 3. : $v_3 = x^2 - \frac{\langle x^2,1 \rangle}{\| \|1\| \|^2} 1 - \frac{\langle x^2,x \rangle}{\| \|x\| \|^2} x = x^2 - \frac{1}{3}$

Thus $\{v_1, v_2, v_3\}$ is an orthogonal basis for $P_2(\mathbb{R})$.

(3) (the normalization process)

$$\|v_1\|^2 = 2, \ \|v_2\|^2 = \frac{2}{3}, \ \|v_3\|^2 = \int_{-1}^{1} \left(t^2 - \frac{1}{3}\right)^2 dt = \left[\frac{t^5}{5} - \frac{2}{9}t^3 + \frac{1}{9}t\right]_{-1}^{1} = 2\left(\frac{1}{5} - \frac{2}{9} + \frac{1}{9}\right) = \frac{8}{45}$$

$$\Rightarrow u_1 = \frac{1}{\sqrt{2}}, \ u_2 = \frac{1}{\|v_2\|} v_2 = \frac{\sqrt{6}}{2} x, \ u_3 = \frac{3\sqrt{5}}{2\sqrt{2}} \left(x^2 - \frac{1}{3}\right) = \frac{\sqrt{10}}{4}(3x^2 - 1)$$

Thus $\{u_1, u_2, u_3\}$ is an orthonormal basis for $P_2(\mathbb{R})$.

◎ The Fourier coefficients

The following result gives us a simple method of representing a vector

as a linear combination of the vectors in an orthonormal basis.

Theorem 6.5. : Let V be an n-dimnsional inner product space. Then

(1) V has an orthonormal basis $\beta = \{v_1, v_2, \ldots, v_n\}$.

(2) For each $x \in V$, it holds that $x = \sum_{i=1}^{n} \langle x, v_i \rangle v_i$.

Proof : (1) Let $\beta_0 = \{w_1, w_2, \ldots, w_n\}$ be an ordered basis for V.

By applying Gram-Schmidt process in Theorem 6.4 to $\beta_0 = \{w_1, w_2, \ldots, w_n\}$,

we obtain an orthogonal set $\beta_1 = \{u_1, u_2, \ldots, u_n\}$ of nonzero vectors such that

$$\text{span}(\beta_1) = \text{span}(\beta_0) = V.$$

By Corollary 2 of Theorem 6.3, we know that β_1 is linearly independent.

In consequence, β_1 is an orthogonal basis for V.

Then $\beta = \{v_1, v_2, \ldots, v_n\}$, where $v_i = \frac{1}{\|u_i\|} u_i$ for $1 \le i \le n$, is an orthonormal basis for V.

(2) See Corollary 1 of Theorem 6.3. ■

Example 6. : In Example 5, we obtain the orthonormal basis $\{u_1, u_2, u_3\}$ for $P_2(\mathbb{R})$ where

$$u_1 = \frac{1}{\sqrt{2}},\ u_2 = \frac{1}{\|v_2\|}v_2 = \frac{\sqrt{6}}{2}x \text{ and } u_3 = \frac{3\sqrt{5}}{2\sqrt{2}}\left(x^2 - \frac{1}{3}\right) = \frac{\sqrt{10}}{4}(3x^2 - 1).$$

Let $f(x) = 1 + 2x + 3x^2$.

Then $\langle f(x), u_1 \rangle = 2\sqrt{2}$,

$$\langle f(x), u_2 \rangle = \frac{2\sqrt{6}}{3} \text{ and}$$

$$\langle f(x), u_3 \rangle = \frac{3\sqrt{10}}{4}\langle f(x), x^2 \rangle - \frac{\sqrt{10}}{4}\langle f(x), 1 \rangle = \frac{3\sqrt{10}}{4}\left(\frac{2}{3} + \frac{6}{5}\right) - \frac{\sqrt{10}}{4} \times 4$$

$$= \frac{7\sqrt{10}}{5} - \sqrt{10} = \frac{2\sqrt{10}}{5}.$$

Therefore we have

$$f(x) = 2\sqrt{2}\,u_1 + \frac{2\sqrt{6}}{3}\,u_2 + \frac{2\sqrt{10}}{5}\,u_3.$$

Corollary : Let V be an n-dimnsional inner product space with an orthonormal basis

$$\beta = \{v_1, v_2, \ldots, v_n\}.$$

Let $T : V \to V$ be a linear operator on V, and let $A = [T]_\beta$.

Then $A_{ij} = \langle T(v_j), v_i \rangle$ for all $1 \leq i, j \leq n$.

Proof : Recall that

$$A = [T]_\beta = \left[[T(v_1)]_\beta \; [T(v_2)]_\beta \cdots [T(v_n)]_\beta \right].$$

By Theorem 6.5, we have

$$T(v_j) = \sum_{i=1}^{n} \langle T(v_j), v_i \rangle v_i \text{ and hence } [T(v_j)]_\beta = \begin{pmatrix} \langle T(v_j), v_1 \rangle \\ \vdots \\ \langle T(v_j), v_n \rangle \end{pmatrix}.$$

In particular, $A_{ij} = \langle T(v_j), v_i \rangle$. ■

Definition : Let V be an (possibly infinite) inner product space with an orthonormal basis β.

Then the scalars $\langle x, y \rangle$ for $y \in \beta$ is called the Fourier coefficients of x relative to β.

◎ The Orthogonal Complement

Definition : Let V be an (possibly infinite) inner product space.

For a nonempty subset S of V, we define $S^{\perp}$ (read "S perp") to be

$$S^{\perp} = \{x \in V \mid \langle x, y \rangle = 0 \text{ for all } y \in S\}.$$

That is, $S^{\perp}$ is the set of all vectors in V that are orthogonal to every vector in S.

The set $S^{\perp}$ is called the orthogonal complement of S.

Remark : (1) $S^{\perp}$ is a subspace of V.

(2) $\{O_V\}^{\perp} = V$ and $V^{\perp} = \{O_V\}$

Example 9. : If $V = \mathbb{R}^3$ and $S = \{e_3\}$, then $S^{\perp}$ equals the xy-plane.

Problem : Let W be the subspace of $\mathbb{R}^3$ defined by $x-2y-5z = 0$, and let $P=(2,5,3) \in \mathbb{R}^3$.

Then find the distance from P to the plane W.

Equivalently, find $Q \in W$ such that the length of $\overline{PQ}$ is minimal.

(1) $v \in W$ is of the form $(2s+5t, s, t)$ for some $s, t \in \mathbb{R}$.

That is, $W = \text{span}(w_1=(2,1,0), w_2=(5,0,1))$.

Note that $W^{\perp} = \{u \in \mathbb{R}^3 \mid \langle u, w_1 \rangle = \langle u, w_2 \rangle = 0\}$.

(2) Let $Q=(2s+5t, s, t) \in W$.

The length of $\overline{PQ}$ is minimal if and only if $\overrightarrow{OP} - \overrightarrow{OQ} = (2-2s-5t, 5-s, 3-t) \in W^{\perp}$.

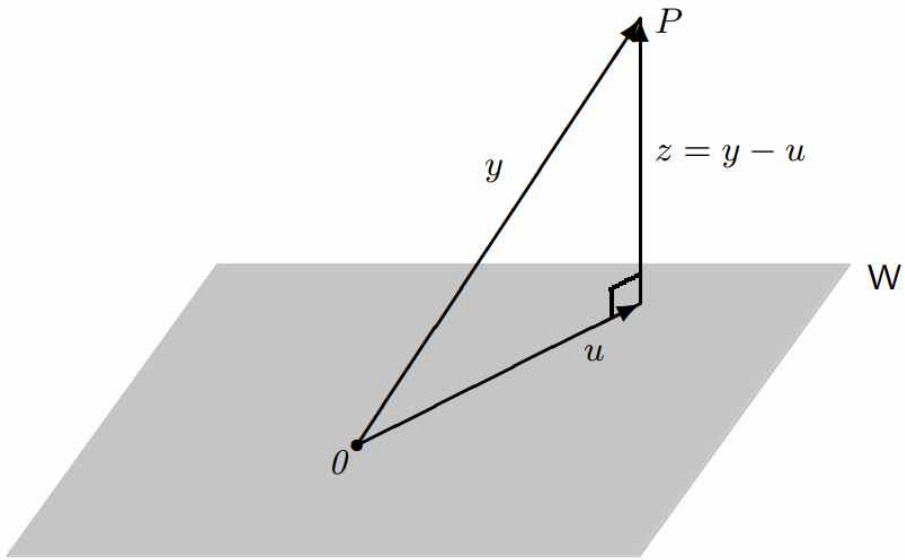
Thus we have

$2(2-2s-5t)+(5-s) = -5s-10t+9 = 0$ and

$5(2-2s-5t)+(3-t) = -10s-26t+13 = 0$

Therefore $(s,t) = \left(\frac{52}{15}, -\frac{5}{6}\right)$ and so

$$Q = \left(\frac{83}{30}, \frac{52}{15}, -\frac{5}{6}\right).$$



Theorem 6.6. : Let V be an inner product space. Let W be a finite dimensional subspace of V. Then for each $y \in V$, there exist unique vectors $u \in W$ and $z \in W^{\perp}$ such that $y = u + z$. Furthermore, if $\{v_1, v_2, \ldots, v_k\}$ is an orthonormal basis for W, then

$$u = \sum_{i=1}^{k} \langle y, v_i \rangle v_i.$$

Some Useful Observations : (1) Let $\{v_1, v_2, \ldots, v_k\}$ be a basis for W. Then

$$W^{\perp} = \{v_1, v_2, \ldots, v_k\}^{\perp} = \{x \in V \mid \langle x, v_1 \rangle = \cdots = \langle x, v_k \rangle = 0\}.$$

(2) $W \cap W^{\perp} = \{O_V\}$.

Proof : (Existence of u and z) Let $\{v_1, v_2, \ldots, v_k\}$ is an orthonormal basis for W.

Also, let $u = \sum_{i=1}^{k} \langle y, v_i \rangle v_i$ and $z = y - u$.

Thus $y = u + z$ and $u \in W$.

Also, $z \in W^{\perp}$ since $\langle z, v_i \rangle = \langle y - u, v_i \rangle = \langle y, v_i \rangle - \langle u, v_i \rangle = 0$.

Theorem 6.6. : Let V be an inner product space. Let W be a finite dimensional subspace of V. Then for each $y \in V$, there exist unique vectors $u \in W$ and $z \in W^{\perp}$ such that $y = u + z$. Furthermore, if $\{v_1, v_2, \ldots, v_k\}$ is an orthonormal basis for W, then

$$u = \sum_{i=1}^{k} \langle y, v_i \rangle v_i.$$

Some Useful Observations : (1) Let $\{v_1, v_2, \ldots, v_k\}$ be a basis for W. Then

$$W^{\perp} = \{v_1, v_2, \ldots, v_k\}^{\perp} = \{x \in V \mid \langle x, v_1 \rangle = \cdots = \langle x, v_k \rangle = 0\}.$$

(2) $W \cap W^{\perp} = \{O_V\}$.

Proof : (Uniqueness of u and z) Suppose that $y = u' + z'$ for some $u' \in W$ and $z' \in W^{\perp}$.

Then $u - u' = z' - z \in W \cap W^{\perp} = \{O_V\}$.

Therefore, $u = u'$ and $z = z'$. ■

Corollary : Let V be an inner product space. Let W be a finite dimensional subspace of V. For a given vector $y \in V$, let $u \in W$ and $z \in W^{\perp}$ be the unique vectors such that $y = u + z$. Then

$\|y - x\| \geq \|y - u\|$ for all $x \in W$, and equality is attained if and only if $x = u$,

and hence u is the unique vector in W that is "closest" to y.

Proof : Let $x \in W$. Then $u - x \in W$ and hence $\langle u - x, z \rangle = 0$.

Thus $\|(u - x) + z\|^2 = \langle (u - x) + z, (u - x) + z \rangle = \|u - x\|^2 + \|z\|^2$.

Therefore, we have

$$\begin{aligned} \|y - x\|^2 = \|u + z - x\|^2 = \|(u - x) + z\|^2 = \|u - x\|^2 + \|z\|^2 \\ \geq \|z\|^2 = \|y - u\|^2. \end{aligned}$$

Moreover, $\|y - x\|^2 = \|z\|^2$ if and only if $\|u - x\|^2 = 0$.

Therefore $\|y - x\| = \|y - u\|$ if and only if $x = u$. ■

Definition : The vector u in the corollary is called the orthogonal projection of y on W.

Problem : Let W be the subspace of $\mathbb{R}^3$ defined by $x-2y-5z = 0$, and let $P=(2,5,3) \in \mathbb{R}^3$.

Then find the distance from P to the plane W.

Equivalently, find $Q \in W$ such that the length of $\overline{PQ}$ is minimal.

(1) $W = \text{span}(w_1=(2,1,0), w_2=(5,0,1))$

Let $u_1=(2,1,0)$ and $u_2 = w_2 - \frac{\langle w_2, u_1 \rangle}{\|u_1\|^2} u_1 = (5,0,1) - \frac{10}{5}(2,1,0) = (1,-2,1)$.

Then $\left\{ v_1 = \frac{1}{\sqrt{5}}(2,1,0), v_2 = \frac{1}{\sqrt{6}}(1,-2,1) \right\}$ is an orthonormal basis for W.

(2) For $y=(2,5,3) \in \mathbb{R}^3$, let

$$u = \langle y, v_1 \rangle v_1 + \langle y, v_2 \rangle v_2 = \frac{9}{5}(2,1,0) + \frac{-5}{6}(1,-2,1) = \left(\frac{83}{30}, \frac{52}{15}, -\frac{5}{6} \right).$$

Corollary to Theorem 6.6 says that u is the unique vector in W that is "closest" to y.

Example 10. : Let $V = P_3(\mathbb{R})$ with the inner product

$$\langle f(x), g(x) \rangle = \int_{-1}^{1} f(t)g(t)dt.$$

Then find the orthogonal projection $f_1(x)$ of $f(x) = x^3$ on $P_2(\mathbb{R})$.

(1) In Example 5, we obtain the orthonormal basis $\{u_1, u_2, u_3\}$ for $P_2(\mathbb{R})$ where

$$u_1 = \frac{1}{\sqrt{2}}, \; u_2 = \frac{1}{\|v_2\|}v_2 = \frac{\sqrt{6}}{2}x \text{ and } u_3 = \frac{3\sqrt{5}}{2\sqrt{2}}\left(x^2 - \frac{1}{3}\right) = \frac{\sqrt{10}}{4}(3x^2 - 1).$$

(2) Theorem 6.6 says that $f_1(x) = \sum_{i=1}^{3} \langle f(x), u_i \rangle u_i$.

Also, we have

$$\langle f(x), u_1 \rangle = \int_{-1}^{1} t^3 \frac{1}{\sqrt{2}}\, dt = 0, \; \langle f(x), u_2 \rangle = \int_{-1}^{1} t^3 \sqrt{\frac{3}{2}}\, t\, dt = \frac{\sqrt{6}}{5}, \text{ and } \langle f(x), u_3 \rangle = \int_{-1}^{1} t^3 \sqrt{\frac{5}{8}}\,(3t^2 - 1)\, dt = 0.$$

Hence

$$f_1(x) = \langle f(x), u_1 \rangle u_1 + \langle f(x), u_2 \rangle u_2 + \langle f(x), u_3 \rangle u_3 = \frac{3}{5}x. \quad \blacklozenge$$

Theorem 6.7. : Let V be an n-dimensional inner product space.

(1) Let $S = \{v_1, v_2, \ldots, v_k\}$ be an orthonormal subset of V. Then

S can be extended to an orthonormal basis $\{v_1, v_2, \ldots, v_k, v_{k+1}, \ldots, v_n\}$ for V.

In this case, if $W = \text{span}(S)$ then $\{v_{k+1}, \ldots, v_n\}$ is an orthonormal basis for $W^{\perp}$.

(2) Let W be a subspace of V. Then $\dim W + \dim W^{\perp} = \dim V$.

Proof : (1) We know already that S can be extended to a basis $\{v_1, v_2, \ldots, v_k, u_{k+1}, \ldots, u_n\}$ for V.

Next, the Gram-Schmidt process gives us

an orthonormal basis $\{v_1, v_2, \ldots, v_k, v_{k+1}, \ldots, v_n\}$ for V.

Note that the first k vectors remain unchanged because

$\{v_1, v_2, \ldots, v_k\}$ is already an orthonormal set.

Proof (Continued) : Next, we will prove that $W^{\perp} = \text{span}(\{v_{k+1},\ldots,v_n\})$.

Obviously, $\{v_{k+1},\ldots,v_n\} \subset W^{\perp}$ and hence $\text{span}(\{v_{k+1},\ldots,v_n\}) \subset W^{\perp}$.

On the other hand, let $y = a_1v_1+\cdots+a_nv_n \in W^{\perp}$.

Then $\langle v_i, y\rangle = a_i = 0$ for all $1 \le i \le k$.

Therefore $y = a_{k+1}v_{k+1}+\cdots+a_nv_n \in \text{span}(\{v_{k+1},\ldots,v_n\})$.

This completes the proof that $W^{\perp} = \text{span}(\{v_{k+1},\ldots,v_n\})$.

Then $\{v_{k+1},\ldots,v_n\}$ is obviously an orthonormal basis for $W^{\perp}$.

(2) It is already shown in (1) that

$$\dim W^{\perp} = |\{v_{k+1},\ldots,v_n\}| = n-k = \dim V - \dim W.$$

Therefore it holds that

$$\dim W + \dim W^{\perp} = \dim V.$$

■

Example : Let $V = \mathbb{R}^3$ and let $w_1 = (a,b,c) \in V$ be such that $a^2+b^2+c^2 = 1$.

(1) Let $W = \text{span}(\{w_1\})$. Then

$$W^{\perp} = \{(x,y,z) \in V \mid ax+by+cz = 0\}.$$

(2) Suppose that $a \neq 0$. If $(x,y,z) \in W^{\perp}$, then

$$(x,y,z) = \left(-\frac{b}{a}y-\frac{c}{a}z, y, z\right) = y\left(-\frac{b}{a},1,0\right)+z\left(-\frac{c}{a},0,1\right).$$

Thus we have $W^{\perp} = \text{span}(\{(b,-a,0),(c,0,-a)\})$.

Now, we apply the Gram-Schmidt process to $\{w_1, w_2 = (b,-a,0), w_3 = (c,0,-a)\}$.

$\Rightarrow v_1 = w_1,\ v_2 = w_2 - \dfrac{\langle w_2, v_1\rangle}{\|v_1\|^2}v_1 = w_2$ and

$$v_3 = w_3 - \frac{\langle w_3, v_1\rangle}{\|v_1\|^2}v_1 - \frac{\langle w_3, v_2\rangle}{\|v_2\|^2}v_2 = (c,0,-a) - \frac{bc}{a^2+b^2}(b,-a,0) = \frac{a}{a^2+b^2}(ac, bc, -(a^2+b^2))$$

$\Rightarrow$ (Normalization process)

$$u_1 = v_1,\ u_2 = \frac{1}{\sqrt{a^2+b^2}}(b,-a,0) \text{ and } u_3 = \frac{1}{\sqrt{a^2+b^2}}(ac, bc, -(a^2+b^2))$$

$\Rightarrow \{u_1, u_2, u_3\}$ is an orthonormal basis for $\mathbb{R}^3$, and $\{u_2, u_3\}$ is an orthonormal basis for $W^{\perp}$.

Exercise #6. :

Let V be an inner product space, and let W be a finite dimensional subspace of V

If $x \notin W$, then there exists $y \in W^{\perp}$ and $\langle x, y \rangle \neq 0$.

Solution : By Theorem 6.6, we can write $x = u + y$ for some $u \in W$ and $y \in W^{\perp}$.

Since $x \notin W$, it follows that $y \neq 0_V$.

Then $\langle x, y \rangle = \langle u + y, y \rangle = \langle u, y \rangle + \langle y, y \rangle = \|y\|^2 > 0$.

Exercise #13. : Let V be an inner product space, and let S and S_0 be subsets of V.

(a) If $S_0 \subset S$, then $S^{\perp} \subset S_0^{\perp}$.

($\because$) If $v \in S^{\perp}$, then $\langle v, x \rangle = 0$ for all $x \in S$.

Thus $\langle v, x \rangle = 0$ for all $x \in S_0$ and hence $v \in S_0^{\perp}$.

(b) $S \subset (S^{\perp})^{\perp}$ and hence $\mathrm{span}(S) \subset (S^{\perp})^{\perp}$.

($\because$) If $v \in S$, then $\langle v, x \rangle = 0$ for all $x \in S^{\perp}$. Thus $x \in (S^{\perp})^{\perp}$.

(c) Let W be a finite dimensional subspace of V. Then $W = (W^{\perp})^{\perp}$.

($\because$) By (b), it holds that $W \subset (W^{\perp})^{\perp}$.

If $x \in (W^{\perp})^{\perp} - W$, then there exists $y \in W^{\perp}$ and $\langle x, y \rangle \neq 0$ (cf. Exercise #6).

But $\langle x, y \rangle = 0$ since $y \in W^{\perp}$ and $x \in (W^{\perp})^{\perp}$.

This is a contradiction.

Thus $(W^{\perp})^{\perp} - W = \varnothing$ and hence $(W^{\perp})^{\perp} \subset W$.

In consequence, $W = (W^{\perp})^{\perp}$.

Exercise #23. : Let $V = \{ (a_1, a_2, \ldots) \mid a_i \in F,\ a_i \neq 0 \text{ for only finitely many } i \}$.

Thus V is an infinite dimensional vector space over F

and $\beta = \{ e_1, e_2, \ldots, e_n, \ldots \}$ is a basis for V.

For $\sigma = (a_1, a_2, \ldots)$ and $\tau = (b_1, b_2, \ldots)$, we define $\langle \sigma, \tau \rangle = \sum_{i=1}^{\infty} a_i \overline{b_i}$.

(a) $\langle \cdot, \cdot \rangle$ is an inner product on V.

(b) $\beta = \{ e_1, e_2, \ldots, e_n, \ldots \}$ is an orthonormal basis for V.

(c) For each $n \geq 2$, let $\sigma_n = e_1 + e_n$. Also, let $W = \text{span}(\{\sigma_n \mid n \geq 2\})$.

Then (i) $e_1 \notin W$ and so $W \neq V$; and (ii) $W^{\perp} = \{0_V\}$.

(c) For each $n \geq 2$, let $\sigma_n = e_1 + e_n$. Also, let $W = \text{span}(\{\sigma_n \mid n \geq 2\})$.

Then (i) $e_1 \notin W$ and so $W \neq V$; and (ii) $W^{\perp} = \{0_V\}$.

Solution : A nonzero element $\sigma \in W$ can be written as

$\sigma = a_1\sigma_{n_1} + \cdots + a_k\sigma_{n_k}$ for some $n_1 < \cdots < n_k$ and nonzero scalars $a_1, \ldots, a_k \in F$.

Then the n_jth entry of σ is equal to a_j. In particular, $\sigma \neq e_1$.

This shows that $e_1 \notin W$ and so $W \neq V$.

Let $\sigma = (a_1, a_2, \ldots)$ be an element of $W^{\perp}$.

Then $\langle \sigma, \sigma_n \rangle = a_1 + a_n = 0$ for all $n \geq 2$.

Thus $\sigma = (a_1, -a_1, -a_1, \ldots, -a_1, \ldots)$.

Now, by the definition of V, it follows that $a_1 = 0$ and hence $\sigma = 0_V$.

Remark : In the above exercise, $(W^{\perp})^{\perp} = V$ and hence $W \neq (W^{\perp})^{\perp}$.

This shows that

the assumption in Exercise 13.(c) that W is finite dimensional is essential.